BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BAHAWALPUR

INTERMEDIATE (12TH CLASS) EXAMINATION, 2024

BIOLOGY

Past Paper 2024

Retyped faithfully from the original board paper — same questions, no changes.

Paper II (Objective Type) — Inter (Ist – A – Exam – 2024)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8467

Note: Four choices A, B, C, D to each question are given. Which choice is correct fill that circle in front of that Question No. on the Objective Bubble Sheet. Use Marker or Pen to fill the circles. Cutting or filling two or more circles will result in Zero Mark in that Question.

Q1. One gram of Ammonia require how much water for its removal :

(A) 300 ml (B) 400 ml

(C) 500 ml (D) 600 ml

Q2. Nociceptor produce the Sensation of ___ :

(A) Pressure (B) Touch

(C) Smell (D) Pain

Q3. Number of Pelvic Vertebrae are :

(A) 9 (B) 12

(C) 7 (D) 5

Q4. The Living cell of Cartilage are called :

(A) Cnidocytes (B) Chondrocytes

(C) Osteocytes (D) Melanocytes

Q5. Hind Brain Contains Medulla, Pons and ___ :

(A) Thalamus (B) Cerebellum

(C) Hypothalamus (D) Cerebrum

Q6. Genetic Code for Methionine is :

(A) AUG (B) UAA

(C) UAG (D) UGA

Q7. Diploid Parthenogenesis occurs in :

(A) Aphid
(B) Honey Bee
(C) Wasp
(D) Hydra

Q8. Study of Aging is called :

(A) Paelontology
(B) Serology
(C) Teratology
(D) Gerontology

Q9. How many different kinds of t.RNA are in Human Cell :

(A) 54
(B) 45
(C) 20
(D) 50

Q10. The Enzyme which joins two pieces of DNA :

(A) Helicase
(B) DNA Ligase
(C) Gyrase
(D) Primase

Q11. Cross which is used to find out the Homozygous or Heterozygous nature of Genotype is called :

(A) Reciprocal Cross
(B) Multiple Cross
(C) Test Cross
(D) Dihybrid Cross

Q12. Mitotic Apparatus is organized during which phase of Mitosis :

(A) Prophase
(B) Metaphase
(C) Anaphase
(D) Telophase

Q13. Antibody used for treatment of Genital Herpes is obtained from :

(A) Rice
(B) Wheat
(C) Soyabean
(D) Corn

Q14. Northern Coniferous Forests are also called :

(A) Savanna
(B) Prairies
(C) Taiga
(D) Tundra

Q15. Which one of following act as Environmental Buffer :

(A) Forests
(B) Desert
(C) Oceans
(D) Lakes

Q16. Endosymbiont Hypothesis was proposed by :

(A) Lynn Margulis
(B) Wallace
(C) Lyell
(D) Darwin

Q17. All Food Chains and Food Webs begin with :

(A) Decomposer Level
(B) Primary Consumer Level
(C) Secondary Consumer Level
(D) Producer Level

Biology (Subjective) — Inter (Ist – A – Exam – 2024)

Time Allowed: 2:40 Hours | **Maximum Marks:** 68

Note: It is compulsory to attempt any (8 – 8) Parts each from Q.No. 2, Q.No.3 and attempt any (6) Parts from Q.No.4. Attempt any (3) Questions from Part – II. Write same Question No. and its Part No. as given in the Question Paper.

Note: Part - I — 22 x 2 = 44 — Make Diagram where necessary.

Q2.

(i) What is Anhydrobiosis?

(ii) Draw a labelled diagram of Flame Cell.

(iii) How Kidney Stones are formed in man?

(iv) Write names of the Bones of Pelvic Girdle.

(v) Define Rickets. Give its cause.

(vi) How do Amoeba move?

(vii) Define Seed Dormancy. How it is beneficial for Plants?

(viii) Mention technique to produce Test Tube Babies.

(ix) Enlist dominant plants of Coniferous Alpine and Boreal Forests.

(x) How Humans affect Tundra Ecosystem?

(xi) How balance in the Nutrient cycle can be upset?

(xii) Acid Rain is Harmful. How?

Q3.

(i) Differentiate between Monohybrids and Dihybrids.

(ii) How pattern of Inheritance of X – Linked Dominant Traits is different from that of X – Linked Recessive Traits?

(iii) What are Linked Genes? Do they obey Law of Independent Assortment?

(iv) Why synthetic Auxins are economical than IAA? Write Commercial uses of 2,4 – D.

(v) Compare Somatic and Autonomic Nervous System.

(vi) What are Pacinian Corpuscles? Write their role as Receptors.

(vii) Define Genomic Library. How can we search a particular Gene from a Genomic Library?

(viii) What are Plasmids? Compare pSC 101 and pBR 322.

(ix) Why patients of SCID are subjected to life threatening infections and how they can be treated via Gene Therapy?

(x) Differentiate between Population and Community.

(xi) How can Grazing turn a Pastureland into a Barren Area?

(xii) What is meant by Denitrification?

Q4.

(i) Differentiate between Growth and Development.

(ii) Define Apical Dominance. How it affects the Morphology of Plants?

(iii) What is Karyotype?

(iv) Do you know how much information one Human Chromosome contains? Comment.

(v) What is Phosphodiester Bond? Give its role in DNA synthesis.

(vi) What is Interphase? Give its Phases.

(vii) Define and draw Mitotic Apparatus.

(viii) Give the role of Ozone in initiation of life on Earth.

(ix) Differentiate between Darwinism and Neo-Darwinism.

> Note: Part - II — (3 x 8 = 24)

Q5.

(a) [4] What is Dialysis? Describe its types.

(b) [4] Describe the importance of Meiosis and Mitosis.

Q6.

(a) [4] Define Secondary Growth. Explain.

(b) [4] Discuss Four Stages of Xerosere.

Q7.

(a) [4] Why Feedback Mechanism is important for a human body? Explain it with an example.

(b) [4] State Hardy – Weinberg Theorem. Discuss any two factors affecting Gene Frequency.

Q8.

(a) [4] Argue that Oogenesis and Spermatogenesis are fundamentally Meiotic Processes but have extraordinary differences.

(b) [4] What is Incomplete Dominance? Explain it with an example.

Q9.

(a) [4] What are Transgenic Plants? Write at least four examples.

(b) [4] Explain the experiment of Haemmerling on importance of Nucleus in development.